



GE VERNOVA

ADMS

16.25.1

Substation File Builder Lite (SFBLite) User Guide

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Change History

Date	Change Description
Aug 25, 2023	ADMS 3.16: • Comprehensive rewrite of document to capture current functionality. • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
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Contents

1. Introduction	4
1.1. SFBLite Workflow	5
2. SFBLite Displays Overview	7
2.1. SFBLite Toolbar and Menus	7
2.2. Build Tab	9
2.3. Configurations Tab	11
2.3.1. Workflow Settings	11
2.3.2. Folders	12
2.3.3. SOAP Adapter	13
2.3.4. Value Placements	14
2.3.5. Build Options	15
2.3.6. SCADA Data Server Connection	16
2.4. Node Assignment Tab	17
2.5. Measurement Mappings Tab	18
2.6. ADMS Measurements	19
2.7. SCADA Devices Tab	21
2.8. SCADA Measurements Tab	22
2.9. Statistics Tab	23
2.10. Log Tab	24
3. Using SFBLite	26
3.1. Starting SFBLite	26
3.2. Command-Line Operations	26
3.3. Building Measurements	27
3.4. Building Value Placements	28
3.5. Calling SFBLite within Model Manager	29

1. Introduction

The Substation File Builder Lite (SFBLite) application is a standalone SCADA mapping application that performs a look-up between the DNOM Network Model and the SCADA Model. When a SCADA Composite Key match is found in the SCADA database, SFBLite gathers all the relevant SCADA Measurements for that device and associates them with the DNOM device. The SFBLite application is intended to be used on projects where the entire network model (both Substation Internals and Externals) come from a single source of truth (such as GIS, Planning, OMS, and so on).

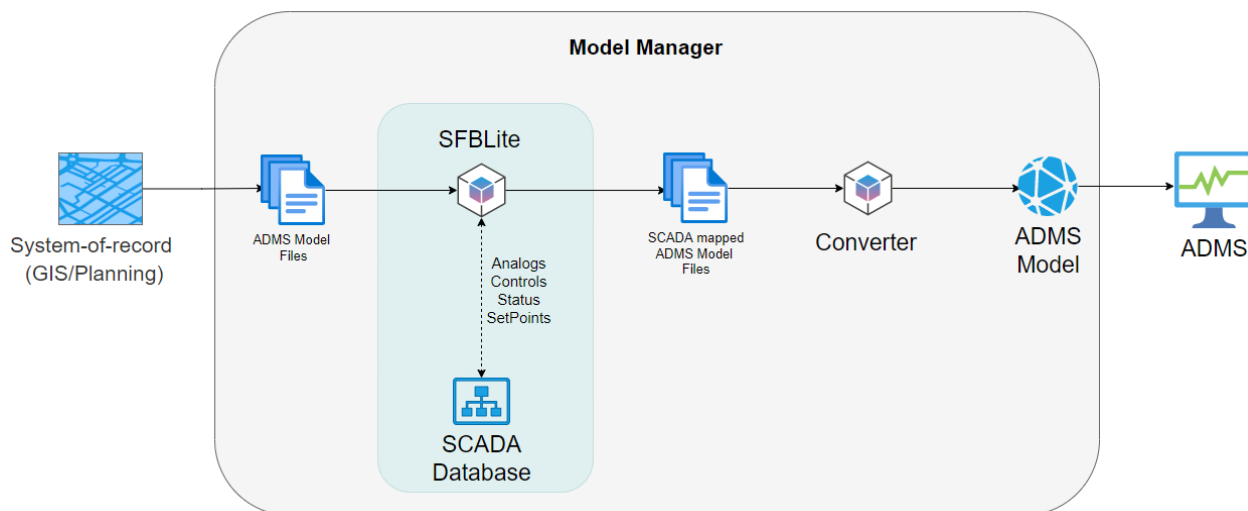


Figure 1. Model Build Process with SFBLite Tool

The SFBLite tool uses the following files as inputs:

- **SFBLite.xml**: SFBLite tool configuration file. This file preserves the modeling support constructs and configuration values that are common to all stations and views. For more details, see [Configurations Tab](#).
- **global.csv**: SFBLite uses this file to find the 8-character <EMSName> in the Substation Externals files to load the devices that exist in the Scada data server.
- **<Station>.xml**: Station Externals files are obtained from the asset management system (typically, GIS or Planning). For more information, refer to the *ADMS Modeling Reference Guide*.
- **<Station>_PLACEMENTS_<View>.xml**: Defines the graphic representation of a given view. SFBLite may use this file to create ValuePlacements if Schematics are available in the placement files.
- **Habitat Database (HDB)**: The SFBLite tool connects to the Habitat database via Sampler to access the Measurement information for every SCADA device. The connection parameters are configured via SFBLite.xml.

Note

To enhance the Station Externals files with the Analog, Status, Control, and Setpoint Measurements, users need to provide ScadaCompositeID attributes for the associated devices.

The building routine uses ScadaCompositeID attributes to extract the existing devices from the HDB server.

To perform the look-up between the DNOM and SCADA model, every SCADA device should have their associated 3-part SCADA Composite Key embedded in the DNOM xml file. In the DNOM file, use ScadaBreakerId on Feeders and ScadaCompositeID on the rest of the DNOM objects. Examples below:

```
<Feeder>
  <Id>d07c5e4c-f722-4624-a227-38048e2a5373</Id>
  <Name>F5635</Name>
  <ScadaBreakerId>3LIONS.FDR.5635</ScadaBreakerId>
```

```
<Switch>
  <Id>97874370</Id>
  <Name>30588</Name>
  <ScadaCompositeID>3LIONS.DASW.30588</ScadaCompositeID>
```

Figure 2. ScadaCompositeID and ScadaBreakerId Examples

Note

If there are multiple SCADA Composite Keys for a single device, additional ScadaBreakerId and ScadaCompositeID definitions (as separate rows) can be written in the DNOM definition. For each field, all associated measurements will be mapped to the DNOM device.

The SFBLite tool outputs the following files:

- **Enhanced <Station>.xml:** All the Substation devices are removed from the <Station>.xml file and are moved to the <Station_internals>.xml file.
- **<Station_internals>.xml:** New Station Internal file includes the Analog, Status, Control, Value Placement, and graphic information from the Scada database along with Substation devices that were originally present in the <Station>.xml file.

1.1. SFBLite Workflow

The SFBLite tool performs the following functions:

- Reads the global.csv file to retrieve the EMS names of Substations used in the HDB.
- Loads the Station Externals files from the specified source folder and extracts ScadaCompositeIDs of devices in the building routine.

- Connects to the HDB server to search for existing Measurement devices with Measurement keys that match with ScadaCompositeID of devices in the input Station Externals files.
- Builds Analogs, Status, Control, and Setpoint Measurements using the assigned ScadaCompositeIDs of devices (both bulk and delta depending on the Configuration). By reading the Point Tag of the Measurement key discovered in HDB, the tool adds the pMeas value to the SCADA Devices.
- Adds appropriate Value Placements to the DNOM devices based on configuration. Value Placements are used to show Analog values on the geographic displays and are not SCADA related.
- Exports the Station External/Internal files with the Analog, Status, Control Measurement mapped to the DNOM devices, along with Value Placements. The tool can be configured to remove all empty attributes in the exported XML files to reduce the file size.

2. SFBLite Displays Overview

This chapter describes the Controls and tabbed displays of the SFBLite tool.

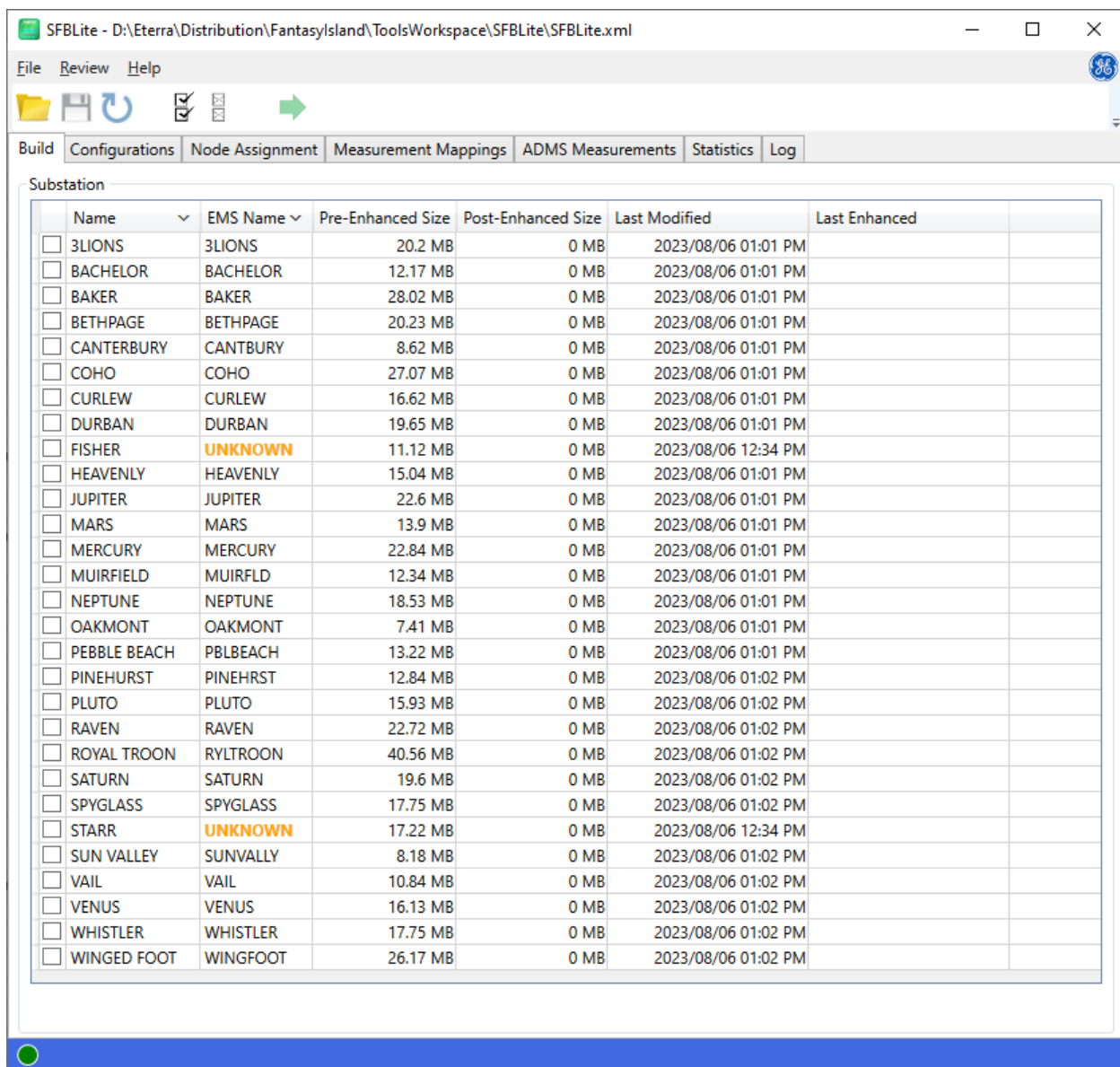


Figure 3. SFBLite Main Display Layout

2.1. SFBLite Toolbar and Menus

The SFBLite tool menus include the following options:

- **File > Open Config...:** Opens a dialog to load the SFBLite configuration file. For more details, see [Configurations Tab](#).
- **File > Reload Configuration:** Reads/reloads the configuration from the SFBLite configuration

xml file.

- **File > Save Config:** Saves the current SFBLite configuration to the xml file.
- **File > Save Config As...:** Provides a File Save option to save the current SFBLite configuration to a new xml file.
- **File > User Preferences:** Opens the xml file that stores the local user's preferred SFBLite dimensions.
- **File > Exit:** Closes the SFBLite application.
- **Review > Export to CSV File:** Provides the option to export the Traced devices (devices that were identified as SCADA devices) and Measurement Keys (SCADA mappings) to CSV file(s).
- **Review > Export to CSV File As:** Provides the option to export the Traced devices (Devices that were identified as SCADA devices) and Measurement Keys (SCADA mappings) to new CSV file(s).
- **Help > User Guide (F1):** Shows the *Substation File Builder Lite User Guide*.
- **Help > About SFBLite:** Shows the SFBLite tool information.

The SFBLite toolbar includes the following buttons:

 **Open:** Opens a dialog to load an SFBLite configuration file.

 **Save Current Configuration:** Saves the current configuration settings.

 **Reload Configuration:** Reloads the SFBLite configuration from the xml file.

 **Select All:** Selects all the stations visible in the grid view.

 **Unselect All:** Unselects all selected stations from the grid view.

 **Perform SCADA Mapping:** Performs SCADA mapping for the selected substations.

The Title bar indicates the path of the currently used SFBLite configuration file (if loaded).

The Status bar at the bottom shows the Scada database connection Status. The color-coded index is given below:

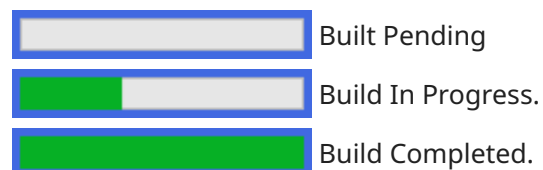
 Connected

 Connection Failure

 Disconnected

The Status bar at the bottom also shows the latest SFBLite Status messages. A detailed log of SFBLite event Statuses can be viewed from the Log Tab.

The build progress bar in the lower-right corner shows the build progress:



2.2. Build Tab

Name	EMS Name	Pre-Enhanced Size	Post-Enhanced Size	Last Modified	Last Enhanced
☒ 3LIONS	3LIONS	20.2 MB	20.2 MB	2023/08/06 01:01 PM	2023/08/08 11:51 AM
☐ BACHELOR	BACHELOR	12.17 MB	0 MB	2023/08/06 01:01 PM	
☐ BAKER	BAKER	28.02 MB	0 MB	2023/08/06 01:01 PM	
☐ BETHPAGE	BETHPAGE	20.23 MB	0 MB	2023/08/06 01:01 PM	
☐ CANTERBURY	CANTBURY	8.62 MB	0 MB	2023/08/06 01:01 PM	
☐ COHO	COHO	27.07 MB	0 MB	2023/08/06 01:01 PM	
☐ CURLEW	CURLEW	16.62 MB	0 MB	2023/08/06 01:01 PM	
☐ DURBAN	DURBAN	19.65 MB	0 MB	2023/08/06 01:01 PM	
☐ FISHER	UNKNOWN	11.12 MB	0 MB	2023/08/06 12:34 PM	
☐ HEAVENLY	HEAVENLY	15.04 MB	0 MB	2023/08/06 01:01 PM	
☐ JUPITER	JUPITER	22.6 MB	0 MB	2023/08/06 01:01 PM	
☐ MARS	MARS	13.9 MB	0 MB	2023/08/06 01:01 PM	
☐ MERCURY	MERCURY	22.84 MB	0 MB	2023/08/06 01:01 PM	
☐ MUIRFIELD	MUIRFLD	12.34 MB	0 MB	2023/08/06 01:01 PM	
☐ NEPTUNE	NEPTUNE	18.53 MB	0 MB	2023/08/06 01:01 PM	
☐ OAKMONT	OAKMONT	7.41 MB	0 MB	2023/08/06 01:01 PM	
☐ PEBBLE BEACH	PBLBEACH	13.22 MB	0 MB	2023/08/06 01:01 PM	
☐ PINEHURST	PINEHRST	12.84 MB	0 MB	2023/08/06 01:02 PM	
☐ PLUTO	PLUTO	15.93 MB	0 MB	2023/08/06 01:02 PM	
☐ RAVEN	RAVEN	22.72 MB	0 MB	2023/08/06 01:02 PM	
☐ ROYAL TROON	RYLTROON	40.56 MB	0 MB	2023/08/06 01:02 PM	
☐ SATURN	SATURN	19.6 MB	0 MB	2023/08/06 01:02 PM	
☐ SPYGLASS	SPYGLASS	17.75 MB	0 MB	2023/08/06 01:02 PM	
☐ STARR	UNKNOWN	17.22 MB	0 MB	2023/08/06 12:34 PM	
☐ SUN VALLEY	SUNVALLY	8.18 MB	0 MB	2023/08/06 01:02 PM	
☐ VAIL	VAIL	10.84 MB	0 MB	2023/08/06 01:02 PM	
☐ VENUS	VENUS	16.13 MB	0 MB	2023/08/06 01:02 PM	
☐ WHISTLER	WHISTLER	17.75 MB	0 MB	2023/08/06 01:02 PM	
☐ WINGED FOOT	WINGFOOT	26.17 MB	0 MB	2023/08/06 01:02 PM	

Figure 4. Build Tab

The Substation Grid includes the following fields and controls:

- **Name:** Station name from the Station Externals file in the SFBLite input folder.
- **EMS Name:** The eight-character station EMS name to link the station to a station in the HDB server.

Note

Defining EMS names in the `global.csv` file is optional to build internals. The tool will search for all EMS names in the `ScadaCompositeIDs` to fetch Measurement keys from HDB server.

For more information about the `global.csv` file, refer to the *ADMS Modeling Reference Tables*.

The following example shows EMS name definitions in `global.csv`:

SubName	0	INDEX	NAME_SUBNAME	EMSMAPNAME_SUBNAME
SubName	1	1	RAVEN	RAVEN
SubName	1	2	SUN VALLEY	SUNVALLY
SubName	1	3	PINEHURST	PINEHRST
SubName	1	4	WHISTLER	WHISTLER
SubName	1	5	MUIRFIELD	MUIRFLD
SubName	1	6	CURLEW	CURLEW
SubName	1	7	OAKMONT	OAKMONT
SubName	1	8	BAKER	BAKER
SubName	1	9	DURBAN	DURBAN
SubName	1	10	ROYAL TROON	RYLTROON
SubName	1	11	3LIONS	3LIONS
SubName	1	12	MERCURY	MERCURY
SubName	1	13	BETHPAGE	BETHPAGE

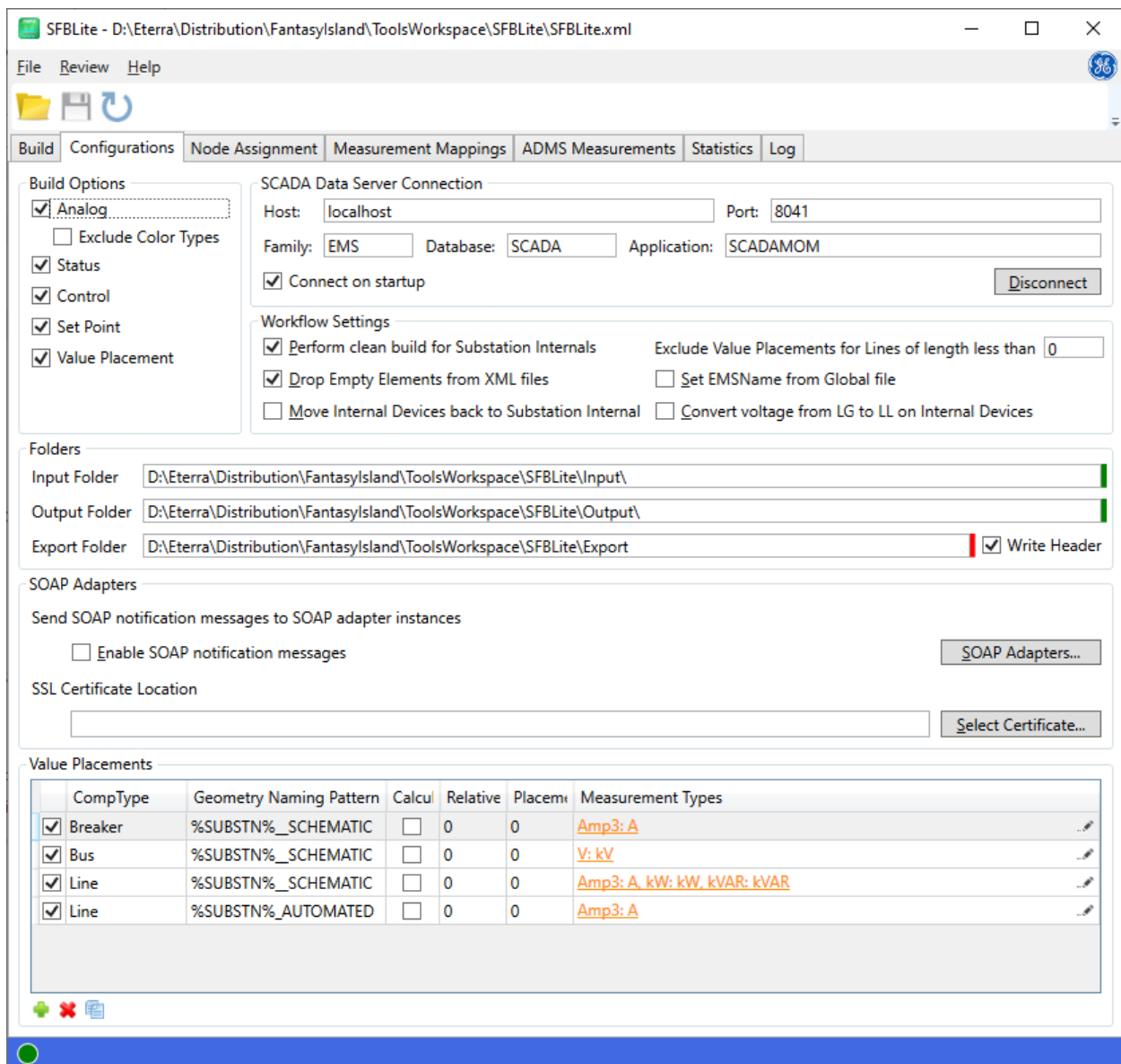
- **Pre-Enhanced Size:** File size before SFBLite was run.
- **Post-Enhanced Size:** File size after SFBLite enhancement.
- **Last Modified:** Last modified (Windows) file timestamp.
- **Last Enhanced:** Last enhanced (through SFBLite) timestamp.
- **Check Selection button:** Checks IsProcessed boxes for all selected station rows.
- **Uncheck Selection button:** Clears IsProcessed boxes for all selected station rows.



Tip

You can use the "Ctrl" or "Shift" key to select multiple station rows. For example, by invoking "Ctrl + A" and then clicking the **Check Selection** button, all stations will be passed to the build routine.

2.3. Configurations Tab



The screenshot shows the SFBLite application window with the 'Configurations' tab selected. The window title is 'SFBLite - D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\SFBLite.xml'. The menu bar includes 'File', 'Review', and 'Help'. The toolbar has icons for file operations. The 'Configurations' tab is active, showing various settings sections.

Build Options

- ☒ Analog
- ☐ Exclude Color Types
- ☒ Status
- ☒ Control
- ☒ Set Point
- ☒ Value Placement

SCADA Data Server Connection

Host: localhost Port: 8041

Family: EMS Database: SCADA Application: SCADAMOM

☒ Connect on startup Disconnect

Workflow Settings

- ☒ Perform clean build for Substation Internals
- ☒ Drop Empty Elements from XML files
- ☐ Move Internal Devices back to Substation Internal
- Exclude Value Placements for Lines of length less than 0
- ☐ Set EMSName from Global file
- ☐ Convert voltage from LG to LL on Internal Devices

Folders

Input Folder: D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Input\

Output Folder: D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\

Export Folder: D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Export ☒ Write Header

SOAP Adapters

Send SOAP notification messages to SOAP adapter instances

☐ Enable SOAP notification messages SOAP Adapters...

SSL Certificate Location

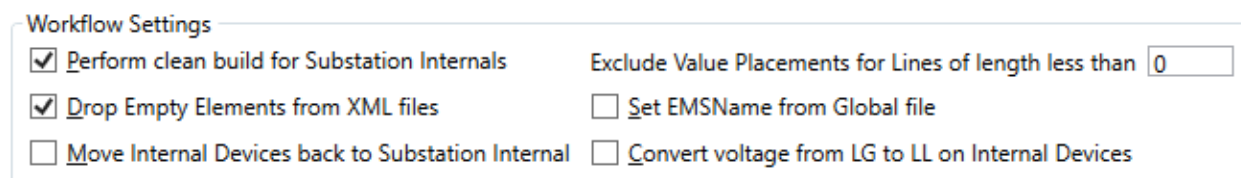
Select Certificate...

Value Placements

	CompType	Geometry Naming Pattern	Calcu	Relative	Placem	Measurement Types
☒	Breaker	%SUBSTN%_SCHEMATIC	☐	0	0	Amp3: A
☒	Bus	%SUBSTN%_SCHEMATIC	☐	0	0	V: kV
☒	Line	%SUBSTN%_SCHEMATIC	☐	0	0	Amp3: A, kW: kW, kVAR: kVAR
☒	Line	%SUBSTN%_AUTOMATED	☐	0	0	Amp3: A

Figure 5. Configurations Tab

2.3.1. Workflow Settings



The screenshot shows the 'Workflow Settings' section of the SFBLite application. It contains several checkboxes and a text input field.

Workflow Settings

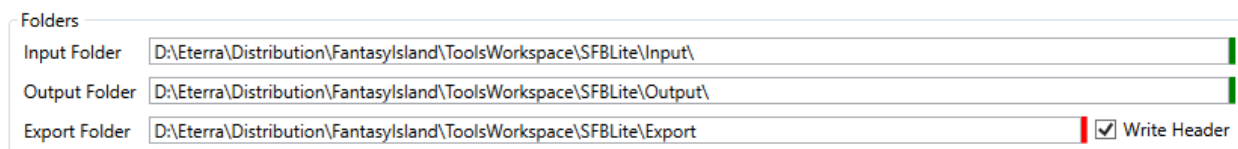
- ☒ Perform clean build for Substation Internals
- ☒ Drop Empty Elements from XML files
- ☐ Move Internal Devices back to Substation Internal
- Exclude Value Placements for Lines of length less than 0
- ☐ Set EMSName from Global file
- ☐ Convert voltage from LG to LL on Internal Devices

Figure 6. Configurations – Workflow Settings

The Workflow Settings group includes the following fields and Controls:

- **Perform Clean Build for Substation Internals:** Specifies whether the tool performs clean build for the Analog, Status, Control, and Setpoint Measurements in the building routine.
 - If selected, the Analog, Status, Control, and Setpoint Measurements that exist in the input externals files are cleared. During the build, the building routine freshly adds the associated Measurements from the HDB database using ScadaCompositeID of the devices.
 - If not selected, the tool traces for changes in the ScadaCompositeID values between the input folder and output folders. The building routine does SCADA mapping for only the new/updated ScadaCompositeID entries.
- **Drop Empty Elements from XML Files:** Enables or disables the clean-up routine for the output XML files. If selected, the tool removes empty DNOM object attributes to reduce the size of the output externals files.
- **Move Internal Device back to Substation Internal:** Moves all Substation devices from the Station External file and keeps them in the Station Internals file. This ensures that many pre-built validations in Converter do not result in validation errors.
- **Set EMSName from Global file:** Read the EMS station file from the global file. This is to correctly set the EMSName if the SCADA station name from GIS is incorrect or unavailable.
- **Convert voltage from LG to LL on Internal Devices:** Typically, LG voltages are set universally in the Station External file. For SFB Lite usage, both Station Internal and External are exported through a single ETL that might set LG voltages across the devices. When Station Internals are created, Converter expects the device voltage to be LL in many validation routines and this flag suppresses those errors.
- **Exclude Value Placements for Lines of length less than:** When Value Placements are enabled, too many Value Placements could get rendered in crowded areas or areas with short line spans. This option provides a mechanism to space out the Value Placements.

2.3.2. Folders



The screenshot shows a configuration window titled "Folders". It contains three text input fields for file paths and a checkbox. The "Input Folder" field contains "D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFB Lite\Input\". The "Output Folder" field contains "D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFB Lite\Output\". The "Export Folder" field contains "D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFB Lite\Export". To the right of the "Export Folder" field is a checkbox labeled "Write Header" which is checked.

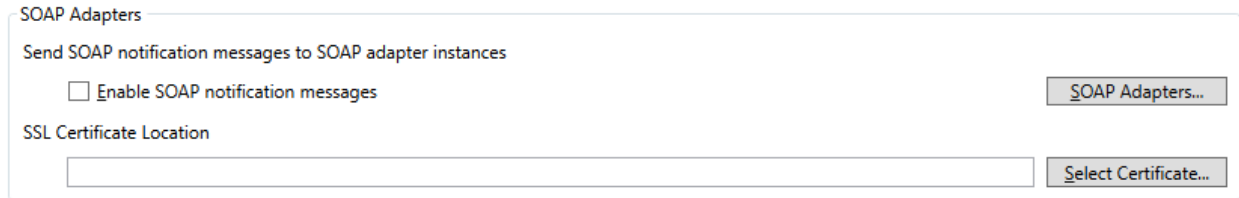
Figure 7. Configurations – Folders

The Folders group includes the following fields and controls:

- **Input Folder:** The path to the folder that stores input Station Externals files.
- **Output Folder:** The path to the folder where the tool exports enhanced Station Externals files when the building routine is completed.
- **Export Folder:** The path to the folder where all manual exports (Mapped Devices, Traced Devices) are stored.

- **Write Header:** Adds the appropriate header to the CSV files (Header + Data).

2.3.3. SOAP Adapter



SOAP Adapters

Send SOAP notification messages to SOAP adapter instances

☐ Enable SOAP notification messages

SSL Certificate Location

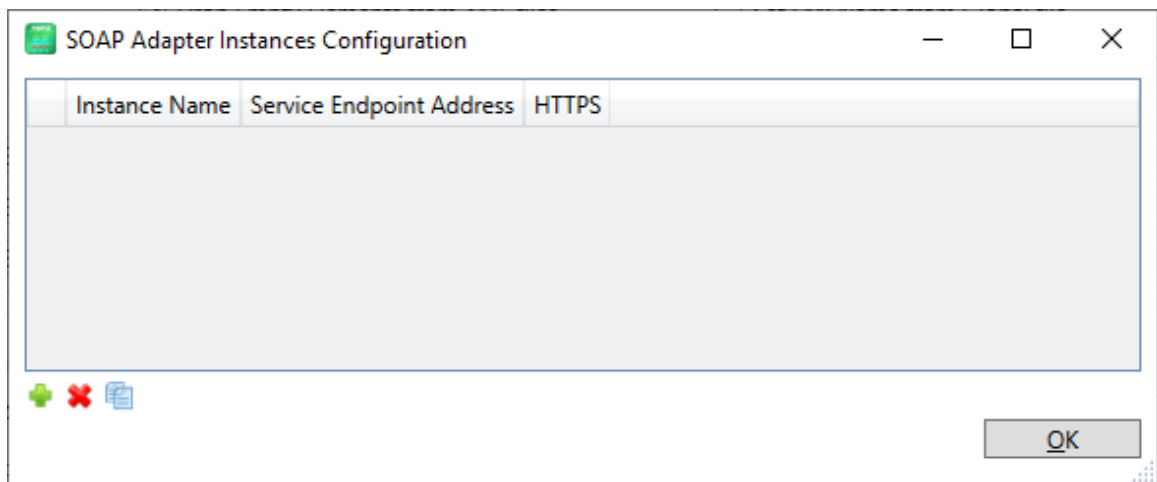
[SOAP Adapters...](#)

[Select Certificate...](#)

Figure 8. Configurations – SOAP Adapter

The SOAP Adapter group includes the following fields and Controls:

- **Enable SOAP Notification Messages:** Enables sending SOAP notification messages from SFBLite to GIS Adapter instances.
 - **SOAP Adapter:** Opens the SOAP Instances Configuration dialog to define names and endpoints for external applications that desire handshake with SFBLite.



SOAP Adapter Instances Configuration

Instance Name	Service Endpoint Address	HTTPS

[+](#) [-](#) [X](#)

[OK](#)

Figure 9. Configurations – SOAP Adapter Instances

- **SSL Certificate Location:** The path to the certificate file for GIS Adapter instances authentication.
 - **Select Certificate:** Opens a dialog to browse to the certificate location.

i Note

It is advisable to use the Model Manager SOAP routine rather than the SFBLite SOAP routine if SFBLite runs within the Model Manager application. This way, redundant SOAP messages can be avoided.

2.3.4. Value Placements

Value Placement attributes can be used for several types of distribution devices (currently supported device types are Breaker, Bus, and Line).

Value Placements						
	CompType	Geometry Naming Pattern	Calculated Placement Rotation	Relative Rotation Angle	Placement Offset	Measurement Types
☒	Breaker	%SUBSTN%_SCHEMATIC	☐	0	0	Amp3: A ..
☒	Bus	%SUBSTN%_SCHEMATIC	☐	0	0	V: kV ..
☒	Line	%SUBSTN%_SCHEMATIC	☐	0	0	Amp3: A, kW: kW, kVAR: kVAR ..
☒	Line	%SUBSTN%_AUTOMATED	☐	0	0	Amp3: A ..

Figure 10. Configurations – Value Placements

The Value Placements toolbar includes the following buttons:

-  **Create:** Creates a new Value Placement group row.
-  **Remove:** Removes the selected Value Placement group row.
-  **Duplicate:** Duplicates the selected Value Placement group row.

The Value Placements grid includes the following attributes:

- **IsEnabled:** Indicates whether the Value Placement group is enabled when the Value Placement building routine is triggered.
- **CompType:** Type of the linked DNOM device object for Value Placement. Currently, the supported distribution device types are Breaker, Bus, and Line.
- **Geometry Naming Pattern:** Geographic geometry type to add Value Placements on. The geometry type follows a certain naming pattern. For example, when the building routine is applied for "3LIONS" Substation, "%SUBSTN%_AUTOMATED" gets resolved as "3LIONS_AUTOMATED" and the Value Placements are added to that declared geometry layer.
- **Measurement Types:** Comma separated list of MeasurementTypes supported by DNOM.
- **Placement Offset:** Distance offset from the original placement of reference object.
- **Relative Rotation Angle:** Incrementally rotates the Value Placement in degree based on the original point placement rotation (Breaker), or line direction (Bus and Line).
- **Calculated Placement Rotation:** If enabled, the text rotation of Value Placement stays fixed. If unchecked, the text rotation follows RelativeRotationAngle.

When the edit button is selected, a Measurement Type popup is pulled up as pictured below. This popup is an alternative way to populate the information. Field descriptions are as follows:

	Measurement Type	Measurement Label	Color Index	Show Arrows
Group 1	Amp3	A	0	False
Group 2	No_MeasType		0	False
Group 3	No_MeasType		0	False
Group 4	No_MeasType		0	False
Group 5	No_MeasType		0	False
Group 6	No_MeasType		0	False

Save Cancel

Figure 11. Configurations – Value Placements Edit Window

- **Measurement Type:** List of all MeasurementTypes supported by DNOM.
- **Measurement Label:** The specified label for that MeasurementType.
- **Color Index:** Index between 0-23 which specifies the ValuePlacement text color.
- **Show Arrows:** When set to **True**, arrows will show which specify the measurement direction.

2.3.5. Build Options

Build Options

☒ Analog

☐ Exclude Color Types

☒ Status

☒ Control

☒ Set Point

☒ Value Placement

Build Internals

Figure 12. Configurations – Build Options

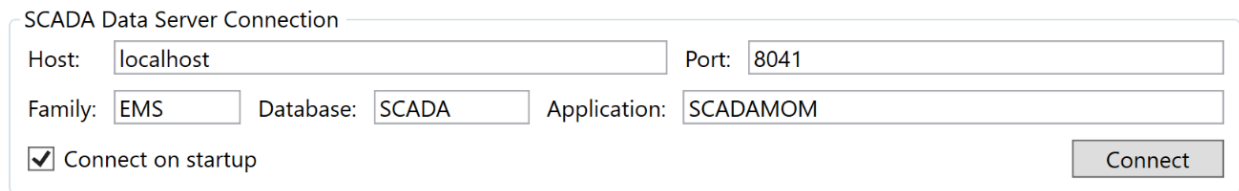
The Build Options group includes the following fields and controls:

- **Analog:** If selected, the tool extracts analog measurement keys from HDB and adds the associated analog measurements to the station externals file.
- **Exclude Color Types:** If selected, users can skip building color measurements that exist in the Scada database.
- **Status:** If selected, the tool extracts status measurement keys from HDB and adds the associated status measurements to the station externals file.
- **Control:** If selected, the tool extracts control measurement keys from HDB and adds the

associated control measurements to the station externals file.

- **Set Point:** If selected, the tool extracts set point keys from HDB and adds the associated set point measurements to the station externals file.
- **Value Placement:** If selected, the tool adds value placements to the station externals file based on the specified value placement groups in the configurations. For more information, see [Value Placements](#).

2.3.6. SCADA Data Server Connection

The image shows a configuration window titled "SCADA Data Server Connection". It contains several input fields: "Host" with the value "localhost", "Port" with the value "8041", "Family" with the value "EMS", "Database" with the value "SCADA", and "Application" with the value "SCADAMOM". There is a checkbox labeled "Connect on startup" which is checked. A "Connect" button is located at the bottom right of the window.

SCADA Data Server Connection

Host: Port:

Family: Database: Application:

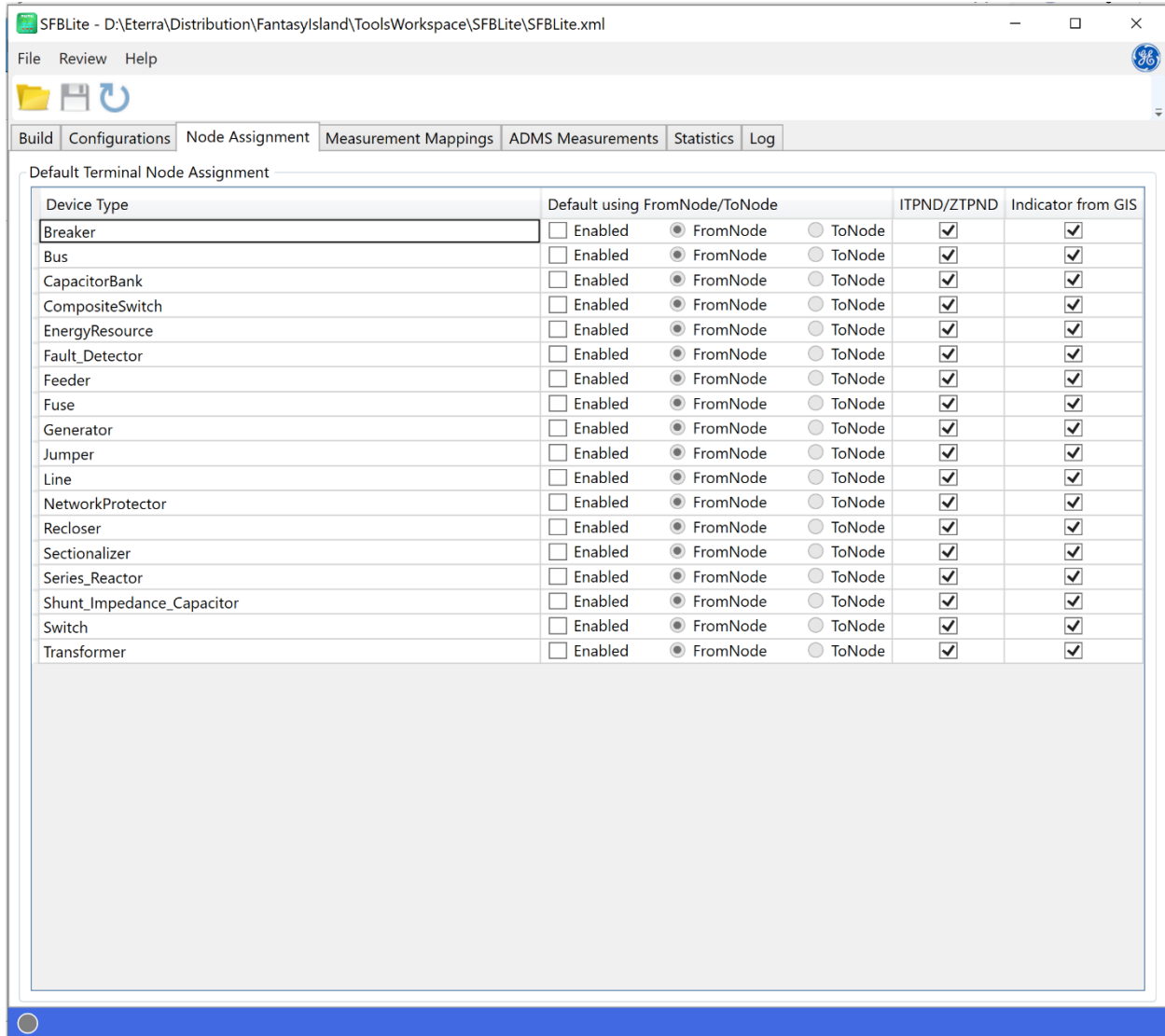
☒ Connect on startup Connect

Figure 13. Configurations – SCADA Data Server Connection Options

The SCADA Data Server Connection group includes the following fields:

- **Host:** Name of the Host for the Scada Data Server.
- **Port:** Port that the SCADA Data Server is connected to.
- **Family:** The corresponding Family name for SCADA (ex. EMS, DMS).
- **Database:** Name of the SCADA Database (ex. SCADA, SCADAMD).
- **Application:** Application context for the SCADA Data Server (ex. SCADA, SCADAMOM).
- **Connect on startup:** If selected, the tool automatically connects to the SCADA Data Server when the application is opened.

2.4. Node Assignment Tab



The screenshot shows the SFBLite application window with the 'Node Assignment' tab selected. The window title is 'SFBLite - D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\SFBLite.xml'. The menu bar includes 'File', 'Review', and 'Help'. The toolbar has icons for file operations. The 'Node Assignment' tab is active, displaying a table titled 'Default Terminal Node Assignment'.

Device Type	Default using FromNode/ToNode	ITPND/ZTPND	Indicator from GIS
Breaker	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Bus	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
CapacitorBank	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
CompositeSwitch	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
EnergyResource	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Fault_Detector	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Feeder	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Fuse	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Generator	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Jumper	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Line	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
NetworkProtector	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Recloser	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Sectionalizer	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Series_Reactor	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Shunt_Impedance_Capacitor	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Switch	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒
Transformer	☐ Enabled ☒ FromNode ☐ ToNode	☒	☒

Figure 14. Node Assignment

This tab contains a table of Measurement mappings. A one-time SCADA-ADMS measurement mapping needs to be done per project to capture all possible SCADA permutations. The table includes the following columns:

- **Device Type:** The DNOM device type such as Switch, Transformer, Fuse, etc.
- **Default using FromNode/ToNode:** When Enabled is checked, the FromNode/ToNode specification will indicate the utilized terminal for all measurements of a certain device type.
- **ITPND/ZTPND:** Checking this option enables the ITPND/ZTPND search to match with measurement devices for the associated device type.
- **Indicator from GIS:** When checked, the mapped SCADA Composite ID value from GIS can be utilized to specify the terminal for the device type. For instance, a Composite ID value of

"3LIONS.TF.T1,1" would specify the ToNode terminal for all T1 transformer measurements. The FromNode terminal is specified by a zero.

2.5. Measurement Mappings Tab

Group	MapName	Type	Device Types	Unit	Phase	Base	Terminal	PictureName	VoltageIsMeasuredLineLine
Analog	AAMP	MeasAmps	(Any)	Base	A	1	0		False
Status	AM	DeviceAutoManual	(Any)	Base	ABC	1	0		False
Analog	AMPA	MeasAmps	(Any)	Base	A	1	0		False
Analog	AMPB	MeasAmps	(Any)	Base	B	1	0		False
Analog	AMPC	MeasAmps	(Any)	Base	C	1	0		False
Analog	AMPG	MeasAmps	(Any)	Base	G	1	0		False
Ctrl	AUTO	SetAutoDeterminationMode	(Any)	Base	ABC	0	0	RO_RECL_CLOSE_DNOM_ONLY	False
Analog	AVLT	MeasVolts	(Any)	Base	A	120	0	ALG_LIM_NS_AVLT	False
Analog	BAMP	MeasAmps	(Any)	Base	B	1	0		False
Analog	BVLT	MeasVolts	(Any)	Base	B	120	0	ALG_LIM_NS_BVLT	False
Analog	CAMP	MeasAmps	(Any)	Base	C	1	0		False
Ctrl	CLOSE	CtrlSwitchStatusCloseOutput	(Any)	Base	ABC	0	0		False
Ctrl	CPOF	CtrlCapStatusOffOutput	(Any)	Base	ABC	0	0		False
Ctrl	CPON	CtrlCapStatusOnOutput	(Any)	Base	ABC	0	0		False
Analog	CVLT	MeasVolts	(Any)	Base	C	120	0	ALG_LIM_NS_CVLT	False
Analog	FA	NoneMeasurementType	(Any)	Base	Unset	0	0		False
Analog	FAA	MeasFaultCurrent	(Any)	Base	A	1	0	ALG_FAULT_CURRENT_RECL	False
Analog	FAB	MeasFaultCurrent	(Any)	Base	B	1	0	ALG_FAULT_CURRENT_RECL	False
Analog	FAC	MeasFaultCurrent	(Any)	Base	C	1	0	ALG_FAULT_CURRENT_RECL	False
Analog	FAG	MeasFaultCurrent	(Any)	Base	G	1	0	ALG_FAULT_CURRENT_RECL	False
Analog	FCLR	FeederColor	(Any)	Base	ABC	0	0		False
Analog	HALO	HaloColor	(Any)	Base	ABC	0	0		False
Analog	KV	MeasVolts	(Any)	Base	ABC	1	0		False
Ctrl	LOWER	CtrlLowerTapOutput	(Any)	Base	ABC	0	0	RO_LTC_LOWER_3P_DNOM_ONLY	False
Status	LR	LtcCapRemoteMeasurementType	(Any)	Base	ABC	0	0		False
Status	LTC	XftapMeasurementType	(Any)	Base	ABC	0	0		False
Ctrl	MANUAL	SetAutoDeterminationMode	(Any)	Base	ABC	0	0		False
Analog	MVA	MeasVltAmps	(Any)	Mega	ABC	1	0	ALG_LIM_NS_MVA	False
Analog	MVAR	MeasVltAmps	(Any)	Mega	ABC	1	0		False

Figure 15. Measurement Mappings

This tab contains a table of Measurement mappings. A one-time SCADA-ADMS measurement mapping needs to be done per project to capture all possible SCADA permutations. The table includes the following columns:

- **Group:** The Measurement mapping type, such as Analog, Status, Control, or Set Point.
- **MapName:** The Map Name that is used in the Measurement keys. This custom value needs to match the SCADA system that is used.
- **Type:** The Measurement type of the Measurement mapping. For the list of supported values, see the *ADMS Modeling Reference Tables*.
- **Device Types:** This option allows a particular measurement to be mapped to "Any" device or a

particular type of device (such as a Breaker only).

- **Unit:** The unit level of the Measurement values, which can be Base, Kilo, or Mega.
- **Phase:** The phase(s) of the Measurement values.
- **Base:** The base value of the Measurement.
- **Terminal:** The terminal to which the measurement gets attached to. "0" is From Terminal, "1" is To Terminal and "2" is Tertiary Terminal.
- **PictureName:** The picture name of the Measurement device used, as defined in the ESET files. This is optional or the same Picture can be re-used across different measurement types.
- **VoltageIsMeasuredLineLine:** If True, the line-to-line voltage is measured. If False, the line-to-ground voltage is measured.
- **CheckPhase:** SFBLite compares the SCADA phase against the DNOM phase and adjusts the measurement accordingly.

For more information on the Measurement mappings, refer to *ADMS Substation Editor File Builder User Guide*.

2.6. ADMS Measurements

Substation Explorer	Substation	ScadaCompositeID	EMS Name	Device Type	Device Name	Device ID	PermitAreaId	DNOM-SCADA Mapping	Mapped	Life Cycle	Mapped Keys	Unmapped Keys
BAKER (06)	BAKER	BAKER.SW.5467F	BAKER	Breaker	5467F	c9e47d8c-7c75-42c	BAKERSW	True	False	Active		
HEAVENLY (3)	BAKER	BAKER.SW.5465F	BAKER	Breaker	5465F	02a61636-6558-49c	BAKERNW	True	False	Active		
	BAKER	BAKER.SW.5464F	BAKER	Breaker	5464F	c1c65116-b7b1-49f	BAKERNW	True	False	Active		
	BAKER	BAKER.SW.5461F	BAKER	Breaker	5461F	a27a3d68-48dd-47f	BAKERNW	True	False	Active		
	BAKER	BAKER.SW.5462F	BAKER	Breaker	5462F	1c0277b5-25c2-4e1	BAKERNW	True	False	Active		
	BAKER	BAKER.SW.5463F	BAKER	Breaker	5463F	0086499a-9861-43f	BAKERSW	True	False	Active		
	BAKER	BAKER.SW.5466F	BAKER	Breaker	5466F	591790a3-da3f-4fc	BAKERNW	True	False	Active		
	BAKER	BAKER.SW.5468F	BAKER	Breaker	5468F	e933049d-e534-47b	BAKERSW	True	False	Active		
	BAKER	BAKER.SW.5469F	BAKER	Breaker	5469F	24c4b224-a822-4b1	BAKERSW	True	False	Active		
	BAKER	BAKER.BUS.23KV_1	BAKER	Bus	23KV_1	f5d0c09-8440-49e	BAKER	True	False	Active		
	BAKER	BAKER.BUS.23KV_2	BAKER	Bus	23KV_2	6d016ee6-905a-46f	BAKER	True	False	Active		
	BAKER	BAKER.BUS.23KV_3	BAKER	Bus	23KV_3	24131e32-be45-4d1	BAKER	True	False	Active		
	BAKER	BAKER.BUS.23KV_4	BAKER	Bus	23KV_4	e3ed516f-84da-466	BAKER	True	False	Active		
	BAKER	BAKER.FUSE.8812	BAKER	Fuse	8812	18fce207-e827-4c6	BAKER	True	True	Active	BAKER.FUSE.8812.STTS BAKER.FUSE.8812.STTS.CLOSE BAKER.FUSE.8813.STTS BAKER.FUSE.8813.STTS.CLOSE BAKER.FUSE.8813.STTS.TRIP	
	BAKER	BAKER.FUSE.8813	BAKER	Fuse	8813	b5f992d2-bdba-46f	BAKER	True	True	Active	BAKER.FUSE.8865.STTS BAKER.FUSE.8865.STTS.CLOSE BAKER.FUSE.8865.STTS.TRIP	
	BAKER	BAKER.FUSE.8865	BAKER	Fuse	8865	25427b22-7be6-49f	BAKER	True	True	Active	BAKER.FUSE.8865.STTS BAKER.FUSE.8865.STTS.CLOSE BAKER.FUSE.8865.STTS.TRIP	
	BAKER	BAKER.SW.3812	BAKER	Switch	3812	4f16b089-1177-4d1	BAKER	True	True	Active	BAKER.SW.3812.STTS	
	BAKER	BAKER.SW.3813	BAKER	Switch	3813	6041b9a4-d84b-48f	BAKER	True	True	Active	BAKER.SW.3813.STTS	
	BAKER	BAKER.SW.21110	BAKER	Switch	21110	473b0c0e-266f-40f	BAKER	True	True	Active	BAKER.SW.21110.STTS	
	BAKER	BAKER.SW.21112	BAKER	Switch	21112	30207e6e-4c04-43b	BAKER	True	True	Active	BAKER.SW.21112.STTS	
	BAKER	BAKER.SW.21111	BAKER	Switch	21111	7e6562aa-cdb0-48f	BAKER	True	True	Active	BAKER.SW.21111.STTS	
	BAKER	BAKER.SW.3865	BAKER	Switch	3865	c1a35aac-853f-4aa	BAKER	True	True	Active	BAKER.SW.3865.STTS	
	BAKER	BAKER.SW.21113	BAKER	Switch	21113	a40cbeff-9007-4d5	BAKER	True	True	Active	BAKER.SW.21113.STTS	
	BAKER	BAKER.SW.5467X	BAKER	Switch	5467X	b95d82ce-3ce3-4cd	BAKER	True	True	Active	BAKER.SW.5467X.STTS	
	BAKER	BAKER.SW.5467Y	BAKER	Switch	5467Y	21c281b8-2b6e-4da	BAKER	True	True	Active	BAKER.SW.5467Y.STTS	
	BAKER	BAKER.SW.5465X	BAKER	Switch	5465X	a682d28e-f165-49e	BAKER	True	True	Active	BAKER.SW.5465X.STTS	

Figure 16. ADMS Measurements

The SFBLite tool traces all the ScadaCompositeIDs in the building routine. This tab shows all the SCADA mapping that was performed by SFBLite. The table includes the following columns:

- **Substation:** The Substation name of the traced device.
- **ScadaCompositeIDs:** The ScadaCompositeID attribute assigned in the Station Externals file.

- **EMS Name:** SCADA Substation name for the particular device/record.
- **Device Type:** The DNOM device type of the traced device.
- **Device Name:** The Device Name of the traced device.
- **Device ID:** The Device ID of the traced device.
- **PermitAreaId:** Permit ID for the traced device.
- **DNOM-SCADA Mapping:** If True, the tool processed the building routine for the specified ScadaCompositeID. If False, the tool found additional devices in the SCADA database that are not specified in the input Station Externals files based on Measurements. Use this to modify or add missing SCADACompositeIDs on the DNOM devices.
- **Mapped:** If True, the tool has added the Measurement devices in the building routing and mapped to the related devices with ScadaCompositeID. If False, the tool failed to find ScadaCompositeID in the HDB and failed to add the Measurement devices in the building routing.
- **Mapped Keys:** List of points present in SCADA database that were successfully mapped.
- **Unmapped Keys:** List of measurements present in SCADA database that were not mapped. Use this list to adjust the "Measurement Mappings" in the configuration till all the ADMS relevant SCADA measurements are added successfully.
- **Life Cycle:**
 - **Unknown:** The life cycle Status of device is yet to be determined.
 - **Active:** The device is active and commissioned.
 - **Proposed:** The device exists, and one or more phases of the devices are planned to be commissioned in the future.
 - **Proposed Remove:** The device exists, and one or more phases of the devices are planned to be de-commissioned in the future.

Note

Users can apply filters to the following columns: Substation, Device Type, DNOM-SCADA Mapping, Mapped, and Life Cycle.

2.7. SCADA Devices Tab

Build	Configurations	Node Assignment	Measurement Mappings	ADMS Devices	Statistics	Log			
Substation Explorer		Substation	ScadaCompositelD	EMS Name	Device Type	Device ID	Mapped	Mapped Keys	Unmapped Keys
BETHPAGE (779)		BETHPAGE	BETHPAGE.CB.4131F	BETHPAGE	CB	4131F	False		BETHPAGE.CB.4131F.DAP BETHPAGE.CB.4131F.FLTA BETHPAGE.CB.4131F.FLTB BETHPAGE.CB.4131F.FLTC BETHPAGE.CB.4131F.FLTE BETHPAGE.CB.4131F.GTE BETHPAGE.CB.4131F.GTE.DISABLE BETHPAGE.CB.4131F.GTE.ENABLE BETHPAGE.CB.4131F.LDSH BETHPAGE.CB.4131F.SORT BETHPAGE.CB.4131F.STTS BETHPAGE.CB.4131F.STTS.CLOSE BETHPAGE.CB.4131F.STTS.TRIP
		BETHPAGE	BETHPAGE.CB.4132F	BETHPAGE	CB	4132F	False		BETHPAGE.CB.4132F.FLTA BETHPAGE.CB.4132F.FLTB BETHPAGE.CB.4132F.FLTC BETHPAGE.CB.4132F.LDSH BETHPAGE.CB.4132F.SORT BETHPAGE.CB.4132F.STTS BETHPAGE.CB.4132F.STTS.CLOSE BETHPAGE.CB.4132F.STTS.TRIP
		BETHPAGE	BETHPAGE.CB.4133F	BETHPAGE	CB	4133F	False		BETHPAGE.CB.4133F.FLTA BETHPAGE.CB.4133F.FLTB BETHPAGE.CB.4133F.FLTC BETHPAGE.CB.4133F.LDSH BETHPAGE.CB.4133F.SORT BETHPAGE.CB.4133F.STTS BETHPAGE.CB.4133F.STTS.CLOSE BETHPAGE.CB.4133F.STTS.TRIP
		BETHPAGE	BETHPAGE.CB.4134F	BETHPAGE	CB	4134F	False		BETHPAGE.CB.4134F.DAP BETHPAGE.CB.4134F.FLTA BETHPAGE.CB.4134F.FLTB BETHPAGE.CB.4134F.FLTC BETHPAGE.CB.4134F.GTE BETHPAGE.CB.4134F.GTE.DISABLE BETHPAGE.CB.4134F.GTE.ENABLE BETHPAGE.CB.4134F.LDSH BETHPAGE.CB.4134F.SORT

Figure 17. SCADA Devices Tab

Note

This tab is hidden by default and is to be used only for debug purposes. Severe performance degradation is to be expected (due to the nature of querying) when using this tab. To expose this tab, remove the `ScadaDevicesReviewTabItem` from the `HideTabs` config option in `SFBLite.xml` file.

```
<HideTabs>ScadaDevicesReviewTabItem,MeasurementKeysReviewTabItem</HideTabs>
```

The SFBLLite tool traces **ALL** the Measurement keys captured from HDB in the building routine. This tab contains a table of Measurement keys returned from the HDB queries and is ordered by SCADA devices available in DNOM model. The table includes the following columns:

- **Substation:** DMS Substation Name
- **ScadaCompositeID:** The Measurement key, which includes the following attributes separated by the dot (.) character.
- **EMS Name:** The eight-character EMS name to link the Substation in the HDB server.
- **Device Type:** The device type that the Measurement key relates to.
- **Device ID:** The device ID that the Measurement key relates to.
- **Mapped:** If True, the tool has added the Measurement devices in the building routing and mapped to the related devices with ScadaCompositeID. If False, the tool failed to find

ScadaCompositeID in the HDB and failed to add the Measurement devices in the building routing.

- **Mapped Keys:** SCADA measurements that were successfully mapped for the device.
- **Unmapped Keys:** SCADA measurements that are present in the SCADA database but did not get mapped for the device.

Note

Users can apply filters to the following columns: EMS Name, Device Type, Device ID, Point, and Mapped.

2.8. SCADA Measurements Tab

SFBLite - D:\Eterra\Distribution\FantasyIsland\Tools\Workspace\SFBLite\SFBLite.xml

File Review Help

Build Configurations Node Assignment Measurement Mappings ADMS Measurements SCADA Measurements Statistics Log

Substation Explorer X

Type	Measurement Key	EMS Name	Device Type	Device ID	Meas	Point	Ctrl	Mapped
Analog	SUBSTN.BETHPAGE.CB.4131F.M.SORT	BETHPAGE	CB	4131F	M	SORT		False
Analog	SUBSTN.BETHPAGE.CB.4132F.M.SORT	BETHPAGE	CB	4132F	M	SORT		False
Analog	SUBSTN.BETHPAGE.CB.4133F.M.SORT	BETHPAGE	CB	4133F	M	SORT		False
Analog	SUBSTN.BETHPAGE.CB.4134F.M.SORT	BETHPAGE	CB	4134F	M	SORT		False
Analog	SUBSTN.BETHPAGE.CB.4135F.M.SORT	BETHPAGE	CB	4135F	M	SORT		False
Analog	SUBSTN.BETHPAGE.CB.4136F.M.SORT	BETHPAGE	CB	4136F	M	SORT		False
Analog	SUBSTN.BETHPAGE.CB.4137F.M.SORT	BETHPAGE	CB	4137F	M	SORT		False
Analog	SUBSTN.BETHPAGE.CB.4138F.M.SORT	BETHPAGE	CB	4138F	M	SORT		False
Analog	SUBSTN.BETHPAGE.CB.4139F.M.SORT	BETHPAGE	CB	4139F	M	SORT		False
Analog	SUBSTN.BETHPAGE.DASW.22222862.M.FAA	BETHPAGE	DASW	22222862	M	FAA		False
Analog	SUBSTN.BETHPAGE.DASW.22222862.M.FAB	BETHPAGE	DASW	22222862	M	FAB		False
Analog	SUBSTN.BETHPAGE.DASW.22222862.M.FAC	BETHPAGE	DASW	22222862	M	FAC		False
Analog	SUBSTN.BETHPAGE.DASW.22222862.M.FAG	BETHPAGE	DASW	22222862	M	FAG		False
Analog	SUBSTN.BETHPAGE.DASW.22224527.M.FA	BETHPAGE	DASW	22224527	M	FA		False
Analog	SUBSTN.BETHPAGE.DASW.22224527.M.FAA	BETHPAGE	DASW	22224527	M	FAA		False
Analog	SUBSTN.BETHPAGE.DASW.22224527.M.FAB	BETHPAGE	DASW	22224527	M	FAB		False
Analog	SUBSTN.BETHPAGE.DASW.22224527.M.FAC	BETHPAGE	DASW	22224527	M	FAC		False
Analog	SUBSTN.BETHPAGE.DASW.22224527.M.FAG	BETHPAGE	DASW	22224527	M	FAG		False
Analog	SUBSTN.BETHPAGE.FDR.4131.M.AAMP	BETHPAGE	FDR	4131	M	AAMP		False
Analog	SUBSTN.BETHPAGE.FDR.4131.M.BAMP	BETHPAGE	FDR	4131	M	BAMP		False
Analog	SUBSTN.BETHPAGE.FDR.4131.M.CAMP	BETHPAGE	FDR	4131	M	CAMP		False
Analog	SUBSTN.BETHPAGE.FDR.4131.M.AVLT	BETHPAGE	FDR	4131	M	AVLT		False
Analog	SUBSTN.BETHPAGE.FDR.4131.M.BVLT	BETHPAGE	FDR	4131	M	BVLT		False
Analog	SUBSTN.BETHPAGE.FDR.4131.M.CVLT	BETHPAGE	FDR	4131	M	CVLT		False
Analog	SUBSTN.BETHPAGE.FDR.4131F.M.FAA	BETHPAGE	FDR	4131F	M	FAA		False
Analog	SUBSTN.BETHPAGE.FDR.4131F.M.FAC	BETHPAGE	FDR	4131F	M	FAC		False
Analog	SUBSTN.BETHPAGE.FDR.4131F.M.FAB	BETHPAGE	FDR	4131F	M	FAB		False
Analog	SUBSTN.BETHPAGE.FDR.4131F.M.FAG	BETHPAGE	FDR	4131F	M	FAG		False
Analog	SUBSTN.BETHPAGE.FDR.4132.M.AAMP	BETHPAGE	FDR	4132	M	AAMP		False
Analog	SUBSTN.BETHPAGE.FDR.4132.M.BAMP	BETHPAGE	FDR	4132	M	BAMP		False
Analog	SUBSTN.BETHPAGE.FDR.4132.M.CAMP	BETHPAGE	FDR	4132	M	CAMP		False
Analog	SUBSTN.BETHPAGE.FDR.4132.M.AVLT	BETHPAGE	FDR	4132	M	AVLT		False

Select All

Display Selected

Display All

Clear

Figure 18. Measurement Keys Review

Note

This tab is hidden by default and is to be used only for debug purposes. Severe performance degradation is to be expected (due to the nature of querying) when using this tab. To expose this tab, remove the MeasurementKeysReviewTabItem from the HideTabs config option in

SFBLite.xml file.

```
<HideTabs>ScadaDevicesReviewTabItem,MeasurementKeysReviewTabItem</HideTabs>
```

The SFBLite tool traces **ALL** the Measurement keys captured from HDB in the building routine. This tab contains a table of Measurement keys returned from the HDB queries. The table includes the following columns:

- **Type:** The Measurement mapping type, such as Analog, Status, Control, or Set Point.
- **Measurement Key:** The Measurement key, which includes the following attributes separated by the dot (.) character.
- **EMS Name:** The eight-character EMS name to link the Substation in the HDB server.
- **Device Type:** The device type that the Measurement key relates to.
- **Device ID:** The device ID that the Measurement key relates to.
- **Point:** The Measurement point name that the Measurement key relates to.
- **Ctrl:** Type of Control associated with the measurement.
- **Mapped:** If True, the tool added the Measurement key to the corresponding device. If False, the tool failed to find the device related in the Station Externals file, and therefore did not map the Measurement key to any device.

Note

Users can apply filters to the following columns: EMS Name, Device Type, Device ID, Point, and Mapped.

2.9. Statistics Tab

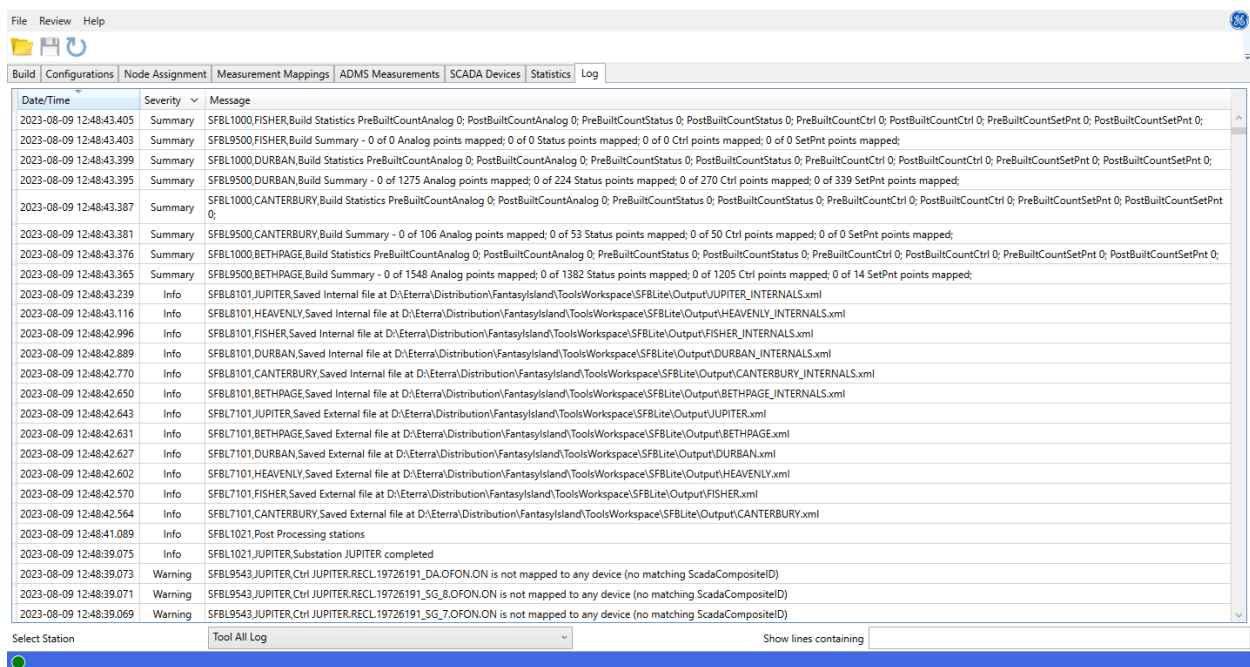
Build	Configurations	Node Assignment	Measurement Mappings	ADMS Measurements	SCADA Devices	Statistics	Log		
Substation									
Name	Last Enhanced	Pre-Built Analog	Pre-Built Status	Pre-Built Ctrl	Pre-Built SetPnt	Post-Built Analog	Post-Built Status	Post-Built Ctrl	Post-Built SetPnt
3LIONS	2023/08/08 11:51 AM	0	0	0	0	0	0	0	0
BACHELOR		0	0	0	0	0	0	0	0
BAKER	2023/08/09 12:40 PM	0	0	0	0	9	82	6	0
BETHPAGE	2023/08/09 12:48 PM	0	0	0	0	0	0	0	0
CANTERBURY	2023/08/09 12:48 PM	0	0	0	0	0	0	0	0
COHO		0	0	0	0	0	0	0	0
CURLEW		0	0	0	0	0	0	0	0
DURBAN	2023/08/09 12:48 PM	0	0	0	0	0	0	0	0
FISHER	2023/08/09 12:48 PM	0	0	0	0	0	0	0	0
HEAVENLY	2023/08/09 12:48 PM	0	0	0	0	75	0	12	0
JUPITER	2023/08/09 12:48 PM	0	0	0	0	0	0	0	0

Figure 19. Viewing Log Messages

The SFBLite tool provides statistics on all measurements that it successfully mapped:

- **Name:** ADMS Substation Name.
- **Last Enhanced:** Timestamp when the Substation was processed through SFBLite.
- **Pre-Build Analog:** Count of Analogs in the original station file.
- **Pre-Build Status:** Count of Status in the original station file.
- **Pre-Build Ctrl:** Count of Controls in the original station file.
- **Pre-Build SetPnt:** Count of SetPoints in the original station file.
- **Post-Build Analog:** Count of Analogs in the enhanced station file.
- **Post-Build Status:** Count of Status in the enhanced station file.
- **Post-Build Ctrl:** Count of Controls in the enhanced station file.
- **Post-Build SetPnt:** Count of SetPoints in the enhanced station file.

2.10. Log Tab



Date/Time	Severity	Message
2023-08-09 12:48:43.405	Summary	SFBL1000.FISHER.Build Statistics PreBuiltCountAnalog 0; PostBuiltCountAnalog 0; PreBuiltCountStatus 0; PostBuiltCountStatus 0; PreBuiltCountCtrl 0; PostBuiltCountCtrl 0; PreBuiltCountSetPnt 0; PostBuiltCountSetPnt 0;
2023-08-09 12:48:43.403	Summary	SFBL9500.FISHER.Build Summary - 0 of 0 Analog points mapped; 0 of 0 Status points mapped; 0 of 0 Ctrl points mapped; 0 of 0 SetPnt points mapped;
2023-08-09 12:48:43.399	Summary	SFBL1000.DURBAN.Build Statistics PreBuiltCountAnalog 0; PostBuiltCountAnalog 0; PreBuiltCountStatus 0; PostBuiltCountStatus 0; PreBuiltCountCtrl 0; PostBuiltCountCtrl 0; PreBuiltCountSetPnt 0; PostBuiltCountSetPnt 0;
2023-08-09 12:48:43.395	Summary	SFBL9500.DURBAN.Build Summary - 0 of 1275 Analog points mapped; 0 of 224 Status points mapped; 0 of 270 Ctrl points mapped; 0 of 339 SetPnt points mapped;
2023-08-09 12:48:43.387	Summary	SFBL1000.CANTERBURY.Build Statistics PreBuiltCountAnalog 0; PostBuiltCountAnalog 0; PreBuiltCountStatus 0; PostBuiltCountStatus 0; PreBuiltCountCtrl 0; PostBuiltCountCtrl 0; PreBuiltCountSetPnt 0; PostBuiltCountSetPnt 0;
2023-08-09 12:48:43.381	Summary	SFBL9500.CANTERBURY.Build Summary - 0 of 106 Analog points mapped; 0 of 53 Status points mapped; 0 of 50 Ctrl points mapped; 0 of 0 SetPnt points mapped;
2023-08-09 12:48:43.376	Summary	SFBL1000.BETHPAGE.Build Statistics PreBuiltCountAnalog 0; PostBuiltCountAnalog 0; PreBuiltCountStatus 0; PostBuiltCountStatus 0; PreBuiltCountCtrl 0; PostBuiltCountCtrl 0; PreBuiltCountSetPnt 0; PostBuiltCountSetPnt 0;
2023-08-09 12:48:43.365	Summary	SFBL9500.BETHPAGE.Build Summary - 0 of 1548 Analog points mapped; 0 of 1382 Status points mapped; 0 of 1205 Ctrl points mapped; 0 of 14 SetPnt points mapped;
2023-08-09 12:48:43.239	Info	SFBL8101.JUPITER.Saved Internal file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\JUPITER_INTERNALS.xml
2023-08-09 12:48:43.116	Info	SFBL8101.HEAVENLY.Saved Internal file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\HEAVENLY_INTERNALS.xml
2023-08-09 12:48:42.996	Info	SFBL8101.FISHER.Saved Internal file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\FISHER_INTERNALS.xml
2023-08-09 12:48:42.889	Info	SFBL8101.DURBAN.Saved Internal file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\DURBAN_INTERNALS.xml
2023-08-09 12:48:42.770	Info	SFBL8101.CANTERBURY.Saved Internal file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\CANTERBURY_INTERNALS.xml
2023-08-09 12:48:42.650	Info	SFBL8101.BETHPAGE.Saved Internal file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\BETHPAGE_INTERNALS.xml
2023-08-09 12:48:42.643	Info	SFBL7101.JUPITER.Saved External file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\JUPITER.xml
2023-08-09 12:48:42.631	Info	SFBL7101.BETHPAGE.Saved External file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\BETHPAGE.xml
2023-08-09 12:48:42.627	Info	SFBL7101.DURBAN.Saved External file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\DURBAN.xml
2023-08-09 12:48:42.602	Info	SFBL7101.HEAVENLY.Saved External file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\HEAVENLY.xml
2023-08-09 12:48:42.570	Info	SFBL7101.FISHER.Saved External file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\FISHER.xml
2023-08-09 12:48:42.564	Info	SFBL7101.CANTERBURY.Saved External file at D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SFBLite\Output\CANTERBURY.xml
2023-08-09 12:48:41.089	Info	SFBL1021.Post Processing stations
2023-08-09 12:48:39.075	Info	SFBL1021.JUPITER.Substation JUPITER completed
2023-08-09 12:48:39.073	Warning	SFBL9543.JUPITER.Ctrl JUPITER.RECL19726191_DA.OFON.ON is not mapped to any device (no matching ScadaCompositeID)
2023-08-09 12:48:39.071	Warning	SFBL9543.JUPITER.Ctrl JUPITER.RECL19726191_SG_8.OFON.ON is not mapped to any device (no matching ScadaCompositeID)
2023-08-09 12:48:39.069	Warning	SFBL9543.JUPITER.Ctrl JUPITER.RECL19726191_SG_7.OFON.ON is not mapped to any device (no matching ScadaCompositeID)

Figure 20. Log Messages

The SFBLite tool provides log messages that the tool encounters during the run. The Log tab contains a table of log messages with the following columns:

- **Date/Time:** The date and time indicates when the log message was created. The time format is "<Year>-<Month>-<Day> <Hour>-<Minute>-<Second>.<Millisecond>".
- **Severity:** Shows the log message severity type.
 - **Info:** Basic information that the tool encounters.
 - **Warning:** Potential issues that may cause failures in routines, such as improper settings.

- **Error:** Errors that the tool encountered during the running process.
- **Summary:** Summary Message.
- **Message:** Shows the description of the log message.

Substation level log messages can be viewed by using the filter available on this screen. The logs are also present in the Output folder.

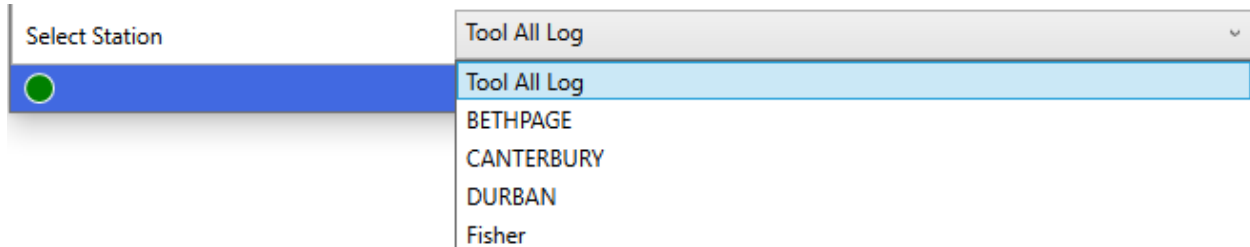


Figure 21. Log Messages – Substation Filter

3. Using SFBLite

3.1. Starting SFBLite

The SFBLite tool is supplied as part of the ADMS Tools kit.

The SFBLite tool requires the path to the SFBLite configuration file as its input parameter. If the configuration file is not already provided in the executable shortcut's Target field, you may need to update the shortcut manually, or open the configuration file when the prompt dialog appears.

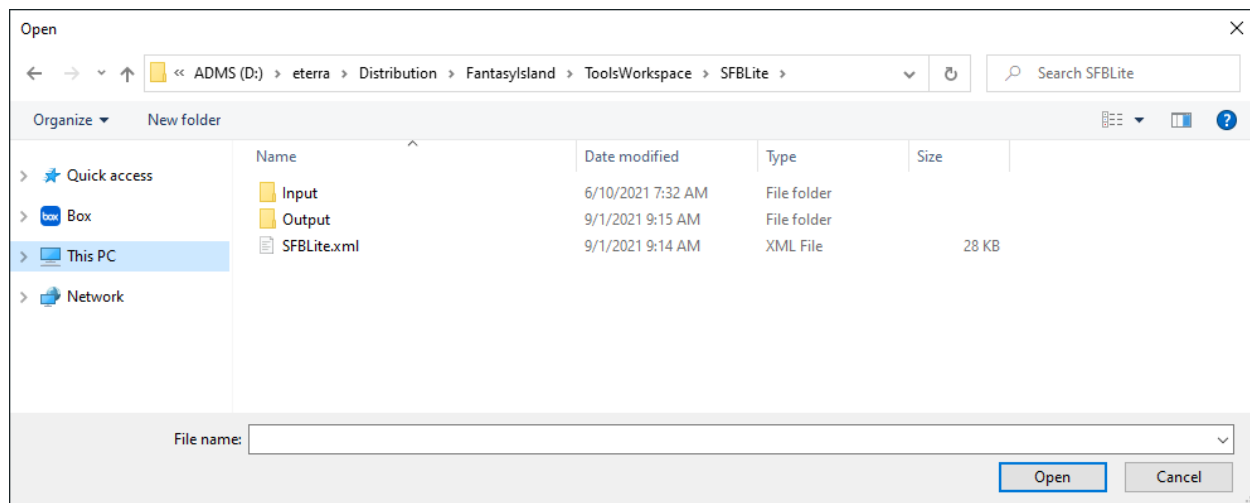


Figure 22. Opening the SFBLite Configuration File

To start the tool:

- From the Windows Start menu, navigate to **Eterra > Tools > Modeling Tools > Substation File Builder Lite**.

The default target value in the shortcut properties dialog box is set as follows:

```
D:\eterra\Distribution\Tools\bin\SFBLite.exe /CONFIG "%IDMS_ROOT%\ToolsWorkspace\SFBLite\SFBLite.xml"
```

3.2. Command-Line Operations

The SFBLite tool can also be run in command-line mode.

Table 1. Command-Line Arguments

Argument	Description
/B	A flag that indicates to run the tool in the batch mode.
/CONFIG [PATH TO FILE]	Path to the SFBLite configuration file.

Argument	Description
/INCLUDE [PATH TO FILE]	Path to the CSV file that contains a list of Substations to be processed by the tool to create the specific Measurement devices or Value Placements. The valid example file contents are: 3LIONS, RAVEN, VENUS or: 3LIONS RAVEN VENUS
/? , /H, /HELP	Displays the user manual of any command or arguments can be used in the tool.
/STATIONS	Name of the Substation to be run by SFBLite. This parameter is optional. If this is not given, SFBLite will process all station files present in its Input folder

The SFBLite tool command-line syntax is as follows:

```
SFBLite.exe /B /CONFIG [PATH TO FILE] /INCLUDE [PATH TO FILE]
```

For example:

```
SFBLite.exe /B /CONFIG %IDMS_ROOT%\ToolsWorkspace\SFBLite\SFBLite.xml /INCLUDE
%IDMS_ROOT%\ToolsWorkspace\SFBLite\SubstationList.csv
```

	A
1	3LIONS
2	RAVEN
3	VENUS

Figure 23. CSV File Example for Running SFBLite in the Batch Mode

✓ Tip

If the INCLUDE path is not specified, the tool searches for all Substation files in the Input folder and builds the internal types as set in the configuration file.

3.3. Building Measurements

The tool allows users to perform a building process for the Analog, Status, Control, and Setpoint Measurement devices. Users can perform the following building routines based on their needs:

- **To perform a clean build for Analog, Status, Control, and Setpoint Measurements:** To perform this operation, select the "Perform clean build for Substation Internals" option in the

Workflow Settings group on the Configurations tab. This building routine initially removes any existing Analog, Status, Control, and Set Point devices from the Station Externals files to perform a clean build, and then creates these Measurements for the devices by ScadaCompositeIDs.

- **To build the Measurement device incrementally:** To perform this operation, clear the "Perform clean build for Substation Internals" option in the Workflow Settings group on the Configurations tab. This building routine preserves the existing Measurement devices in Station Externals files in the output folder and updates the Measurement devices based on the incoming ScadaCompositeID attributes.

If the "Remove empty elements in the output XML files" option in the Workflow Settings group of the Configuration tab is selected, the tool removes all empty XML elements from the output Station Externals files.

To build the Analog, Status, Control, and Setpoint Measurement devices:

1. In the Station Externals group of the Build tab, select the Station Externals to build.
2. In the Build Options group of the Build tab, select the desired types of Measurement devices (Analog, Status, Control, or Set Point) to retrieve from the HDB database.
3. Click the Build Internals button to initiate the building routine.

When the building routine is completed, the Status bar message shows the build Status, such as: "Substation <StationName> is saved in the output folder."

The devices of the introduced Measurement devices are added to the Traced Devices Review tab. The Measurement Keys Review tab shows the full Measurement keys found in the HDB database, with mapping Statuses for the devices corresponding to the ScadaCompositeIDs.

3.4. Building Value Placements

The tool can perform a building process for Value Placements using the Value Placement group configuration that is defined on the Configurations tab. The following table shows Value Placement conditions for different device types.

Table 2. Placement Conditions of Value Placements

CompType	Rotation	Placement Location
Breaker	Follows the settings of Value Placement group. If is <TextRotationCalculated> set to false, it follows the degree attribute of <RelativeRotationAngle>.	At the location of the component's geometry placement.
Bus	N/A	On the last vertex of the geometry placement.
Line	N/A	At the middle point of the whole line segment. The Value Placement is aligned with the direction of the line.

To build Value Placements:

1. Configure the Value Placement groups. You must provide valid attributes, as described in [Value Placements](#). In Geometry Naming Pattern (GeometryType), the %SUBSTN% symbol can be resolved to represent <StationName> for the corresponding Substations in the build.
2. In the Station Externals group of the Build tab, select the Station Externals to build.
3. In the Build Options group of the Build tab, select the **Value Placement** option.
4. Click the **Build Internals** button to initiate the building routine.

When the building routine is completed, the Status bar message shows the build Status, such as: "Substation <StationName> is saved in the output folder."

If the "Remove empty elements in the output XML files" option in the Workflow Settings group of the Configurations tab is checked, the tool removes all empty XML elements from the output Station Externals files.

Note

Performing a clean build is not applicable to the Value Placement building routine. The tool always preserves all existing Value Placement entries in the input files.

3.5. Calling SFBLite within Model Manager

Like other Modeling tools that can be invoked within Model Manager application, SFBLite can be used to enhance the stations as they get processed within Model Manager. This can only be done if the Model Manager (which typically resides in DMZ) can make a call to the SCADA server (which resides in Protected zone) through Sampler. If the modeling architecture is different in the project, invoke SFBLite separately in an environment where SCADA database is available.

To invoke SFBLite, please create a PowerShell script that calls the command-line arguments described above. A sample PowerShell script is given below:

```
# Notes on how to change the PowerShell execution policy. Out of the box the PowerShell execution
policy will be 'Restricted' for
# all users.
# To see your current (user) execution policy:
#>PowerShell -Command "& \get-executionpolicy"

# To set the execution policy for the CurrentUser to Unrestricted:
#>PowerShell -Command "& \{Set-Executionpolicy Unrestricted -scope CurrentUser}"

# To set the execution policy for all users on this machine to Unrestricted. Note this requires
running from an Administrator enabled command window:
#>PowerShell -Command "& \{Set-Executionpolicy Unrestricted -scope LocalMachine}"

$stationNames = $null

$scriptpath = $MyInvocation.MyCommand.Path
```

```

$scriptdir = Split-Path $scriptpath

for ($i=0; $i -lt $args.Count; $i++)

{ if ($args[$i] -eq "-SSG" -and $args[$i+1].InputType -eq "Externals")
    {
        $substationsList = $args[$i+1].Substations
        for ($j=0; $j -lt $substationsList.Count; $j++)
        {
            $stationName = $substationsList[$j].Name
        }
    }
}

$process = New-Object System.Diagnostics.Process
$process.StartInfo.UseShellExecute = $false
$process.StartInfo.RedirectStandardOutput = $true
$process.StartInfo.RedirectStandardError = $true
$process.StartInfo.CreateNoWindow = $true

$process.StartInfo.FileName = $env:DMS_TOOLS_BINDIR + "SFBLite.exe"
$process.StartInfo.Arguments = "/B /CONFIG %IDMS_ROOT%\ToolsWorkspace\SFBLite\SFBLite.xml /STATIONS
    $stationName"

# Log the command line
"SFBLite started with the command:" | Out-String
$process.StartInfo.FileName + " " + $process.StartInfo.Arguments | Out-String
$process.Start() | Out-Null
$process.StandardOutput.ReadToEnd() | Out-String
$process.StandardError.ReadToEnd() | Out-String
$process.WaitForExit();

$host.SetShouldExit(0)

```

It is advised to finish the SCADA mapping before Converter runs within Model Manager. To achieve this, run the PowerShell script at the "Converting" step with the "Synchronous" flag set to true.

 Edit Script Object X

Name:

Hint:

Enable: ☒

Synchronous: ☒ (If checked, Model manager backend workflow will be blocked while this script is running.)

Timeout Limit:  Seconds

Trigger: Conversion Status ▾

☐ Publishing	☐ ConverterFataIs	☐ SSGError	☐ Reviewi
☒ Converting	☐ ErrorCopyInputToProcessing	☐ UnableToRestore	☐ ModelM
☐ VerifyingInputData	☐ PublishingError	☐ ConvertingError	☐ Modeli
☐ MovingToProcessing	☐ Uploaded	☐ SchemaValidationFailed	
☐ Queued	☐ Timeout	☐ Removed	
☐ ReadyForReview	☐ Rejected	☐ UnderReview	

☐ Enable triggering Scripts on MMA Idle State.

Script Trigger - Idle Delay (after 'X' seconds idle)



PowerShell Script 

OK Cancel

Figure 24. Model Manager Script Options Window